

H.120

The first video compression standard was developed in 1984, as an attempt to standardise file encoding methods

How stuff works

neous information regarding the file can be stored on the file itself. This information includes the title, album name, artist and date and are useful in identifying audio tracks.

In older audio CDs, 3-4 minute tracks were about 60 MB in size. These were uncompressed audio files stored losslessly. When you use a CD ripper to convert this into an MP3 file, certain sections of the audio are removed. The MP3 standard, does not specify an exact algorithm for converting RAW audio into the final MP3 file. It however, gives extensive information on what sort of audio signals are inaudible / unperceivable by the human ear and can be removed. Based on research in psychoacoustics (the study of how sound is perceived by the human mind), it has also generated a list of the types of audio signals which even though physically perceived by the human ear, is still ignored by the brain.

Different types of MP3 encoders use this information, to remove these types of redundant/useless signals and then convert the file. The following signals are removed:

1. Signals with frequencies above and beyond the auditory spectrum (less than 20 samples per second and more than 20,000 samples per second).
2. Low amplitude signals (of any frequency) in the presence of certain types of signals of much higher amplitudes.

Codecs remove fractions of the original file. While some remove only the most inaudible of the sounds, leading to an output MP3 that is still considerably large, some remove a sizeable portion of the whole file, retaining only the signals deemed most essential for the listening experience, and produce much smaller files. Most codecs allow you to make this decision regarding what quality to retain, through the concept of bit rates.

Bit Rates

Most codecs used to rip CDs into MP3 files allow you to select a bit rate. This lets you control how much of the original file will be removed, as well as the quality and file size of the MP3 file. A 128 kbps

OTHER FORMATS

In addition to MP3, some of the other compression formats are:

MP3PRO:

A combination of MP3 with a technique known as spectral band amplification. Here, sounds in different frequencies are encoded separately, MP3PRO offers comparable sound quality at file sizes often half of that in MP3.

OggVorbis:

This open source, free compression mechanism produces similar quality to MP3 at much lower bitrates and file sizes. Windows Media Audio (WMA):

A competing proprietary technology developed by Microsoft, WMA offers different codecs for different applications such as music and voice. It also has support for digital rights management, allowing an online vendor to prevent copy and redistribution of a track. This is a crucial technology that is lacking in MP3.

While each codec has its own advantages and disadvantages, current popularity, support on personal players and widespread usage has ensured that none will compete with MP3 for a long time to come.

MP3 file will use 128 kilo bits of data for every second of playback. Thus, the size of the file (in MB) will be $\text{Bitrate (in kbps)} \times \text{length of the file (in seconds)} / 8 \times 1024$.

A bit rate of 128 kbps is often high enough to make the final MP3 file virtually indistinguishable from a CD version of the same song and is hence popular. Lower bitrates often lead to perceivable differences in quality in high end speakers.

While you larger bitrates to maintain quality on a complex file with a large amount of randomness, single-instrument audio files, often retain the original quality even at lower bit rates.

While earlier MP3s used the same bit rate throughout the file, more recent ones have started using the concept of variable bit rate (VBR) where a different bit rate is used depending upon the types of sounds in each frame of the MP3 file, leading to much higher efficiency in compression while maintaining the quality of the file. The downside is that the file size is lesser than in the case of constant bit rate. ■

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